WORKED SOLUTION

STEP III 2009, Question 7

Pure • Repeated Differentiation and Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q7

The key idea

Everything flows from one innocent identity: $(1+x^2)f_0(x) = 1$. Differentiating it repeatedly is what generates relations between consecutive derivatives, and a relation that holds for one n can be differentiated once more to give the next one, which is exactly the shape of an induction step. In part (ii), multiplying by the right power of $1+x^2$ clears every denominator, and a second induction tracks the leading term to pin down the degree.

Solution

Part (i)

Write f_n for $f_n(x)$ and $P(n)$ for the statement

$$(1+x^2)f_{n+1} + 2(n+1)xf_n + n(n+1)f_{n-1} = 0.$$

Base case. Since $(1+x^2)f_0 = 1$ for all x, differentiating (with $f_0' = f_1$) gives

$$(1+x^2)f_1 + 2xf_0 = 0.$$

Differentiating again, using the product rule on each term,

$$(1+x^2)f_2 + 2xf_1 + 2xf_1 + 2f_0 = 0, \qquad \text{that is,} \qquad (1+x^2)f_2 + 4xf_1 + 2f_0 = 0,$$

which is $P(1)$.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$, for all x. Since it is an identity in x, we may differentiate it. Using $f_k' = f_{k+1}$,

$$\begin{aligned} 0 &= (1+x^2)f_{n+2} + 2xf_{n+1} + 2(n+1)f_n + 2(n+1)xf_{n+1} + n(n+1)f_n \\ &= (1+x^2)f_{n+2} + 2(n+2)xf_{n+1} + (n+1)(n+2)f_n, \end{aligned}$$

since $2+2(n+1) = 2(n+2)$ and $2(n+1)+n(n+1) = (n+1)(n+2)$. This is $P(n+1)$. So by induction $P(n)$ holds for all $n \geq 1$, as required.

(Equivalently, apply Leibniz's rule to differentiate $(1+x^2)f_0 = 1$ exactly $n+1$ times: only three terms survive, because the third derivative of $1+x^2$ is zero, and they are the three terms of $P(n)$.)

Part (ii)

The first three. We have $P_0 = (1+x^2)f_0 = 1$. Next,

$$f_1 = \frac{-2x}{(1+x^2)^2}, \qquad \text{so} \qquad P_1 = (1+x^2)^2 f_1 = -2x.$$

Differentiating f_1 by the quotient rule,

$$f_2 = \frac{-2(1+x^2)^2 + 2x \cdot 2(1+x^2)\cdot 2x}{(1+x^2)^4} = \frac{-2(1+x^2)+8x^2}{(1+x^2)^3} = \frac{6x^2-2}{(1+x^2)^3},$$

so $P_2 = (1+x^2)^3 f_2 = 6x^2 - 2$. (Part (i) with $n=1$ gives the same: $(1+x^2)f_2 = -4xf_1 - 2f_0$.)

The recurrence. Differentiate $P_n = (1+x^2)^{n+1} f_n$ by the product rule:

$$P_n' = 2(n+1)x(1+x^2)^n f_n + (1+x^2)^{n+1} f_{n+1}.$$

Multiplying through by $1+x^2$ turns each term into a P:

$$(1+x^2)P_n' = 2(n+1)x\,(1+x^2)^{n+1} f_n + (1+x^2)^{n+2} f_{n+1} = 2(n+1)xP_n + P_{n+1}.$$

Rearranging,

$$P_{n+1} - (1+x^2)P_n' + 2(n+1)xP_n = 0$$

for every $n \geq 0$, as required. No induction is needed here; it is a direct calculation valid for each n.

> **Ask yourself** How do we know the degree is exactly n, and not smaller because the top terms cancel?

We prove a slightly stronger statement, which carries the information we need through the induction: *P_n is a polynomial of degree n with leading coefficient $a_n \neq 0$.* The case $n=0$ holds, since $P_0 = 1$.

Suppose it holds for some $n \geq 0$, so $P_n = a_n x^n + (\text{lower powers})$ with $a_n \neq 0$. By the recurrence,

$$P_{n+1} = (1+x^2)P_n' - 2(n+1)xP_n.$$

The right-hand side is a polynomial, because P_n and so P_n' are. Now $P_n' = na_n x^{n-1} + \cdots$ (this is 0 when $n=0$, which matches $na_n = 0$), so $(1+x^2)P_n'$ has degree at most $n+1$ with x^{n+1} coefficient na_n. Also $2(n+1)xP_n$ has degree $n+1$ with leading coefficient $2(n+1)a_n$. Hence P_{n+1} has degree at most $n+1$, and its x^{n+1} coefficient is

$$a_{n+1} = na_n - 2(n+1)a_n = -(n+2)a_n.$$

This is nonzero, because $a_n \neq 0$ and $n+2 > 0$. So the top terms cannot cancel, and P_{n+1} has degree exactly $n+1$. By induction, P_n is a polynomial of degree n for every $n \geq 0$.

As a bonus, $a_{n+1} = -(n+2)a_n$ with $a_0 = 1$ gives $a_n = (-1)^n(n+1)!$, matching $a_1 = -2$ and $a_2 = 6$ above.

Checks

- Part (i) at $x = 0$, $n = 1$: from P_n we get $f_0(0) = 1$, $f_1(0) = 0$, $f_2(0) = -2$, and $f_2 + 0 + 2f_0 = -2 + 2 = 0$.
- The recurrence with $n = 1$: $(1+x^2)P_1' - 4xP_1 = -2 - 2x^2 + 8x^2 = 6x^2 - 2 = P_2$.
- Numerically at $x = 0.7$, a central difference gives $f_0''(0.7) \approx 0.284164$, and $(6x^2 - 2)/(1+x^2)^3 = 0.284164$. We also checked part (i) and the recurrence exactly in rational arithmetic up to $n = 12$ at 200 random rational points.

Common slips

- Not proving a base case that matches the stated range: the identity in part (i) starts at $n = 1$, so the base case must be $n = 1$, which needs f_2.
- Saying "the degree goes up by one each time" without checking that the x^{n+1} terms of $(1+x^2)P_n'$ and $2(n+1)xP_n$ don't cancel. They don't, because $n \neq 2(n+1)$, but it has to be said.

Beyond STEP

Splitting into partial fractions with complex numbers, $\frac{1}{1+x^2} = \frac{1}{2\mathrm{i}}\left(\frac{1}{x-\mathrm{i}} - \frac{1}{x+\mathrm{i}}\right)$, makes every derivative easy and gives the closed form $P_n(x) = (-1)^n\, n!\, \operatorname{Im}(x+\mathrm{i})^{n+1}$ (try it for $n = 1, 2$). Writing $x = \cot\theta$ then shows that the zeros of P_n are $\cot\frac{k\pi}{n+1}$ for $k = 1, \ldots, n$: all real and distinct, like the zeros of the classical orthogonal polynomials, which are also generated by repeatedly differentiating a simple function. Needham's *Visual Complex Analysis* is a lovely place to see why complex numbers tame calculations like this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.